Practice workbook · Name the food-test reagent or read food-test results

One-step food-test questions: naming the reagent or equipment for a test, or reading colour outcomes to decide which nutrients each sample holds.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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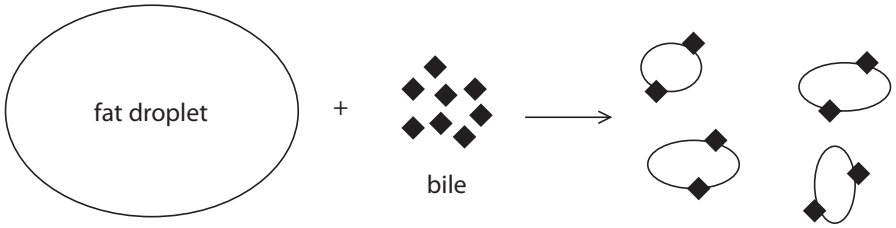
Type ID: HBI-FOOD-TEST-INTERPRET

1. Question 3 · part b.i · 3 marks

January 2023 Paper 1

Worked-example retry · Attempt part b.i only.

3 (a) The diagram represents the effect of bile on a large fat droplet.



(i) Where is bile produced?

(1)

- ☒ **A** gall bladder
- ☒ **B** liver
- ☒ **C** small intestine
- ☒ **D** stomach

(ii) Describe the role of bile in the digestion of fat.

(3)

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1. Question 3 · part b.i · 3 marks

January 2023 Paper 1

Worked-example retry · Attempt part b.i only.

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(b) Three food tests are used on a solution of food.

The table shows the reagent used and the colour of the food solution after each test.

Reagent used	Colour of the food solution after the test
iodine solution	blue/black
biuret solution	purple
Benedict's solution	blue

(i) Describe what the food tests show about the dietary components in the food.

(3)

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(ii) Saliva is added to a new sample of the food solution and the mixture is kept at 37 °C for 10 minutes.

The food tests are then repeated and the solution turns brick red when tested with Benedict's reagent.

Explain the effect of saliva on the food solution.

(3)

1. Question 3 · part b.i · 3 marks · Working and answer

Working and answer

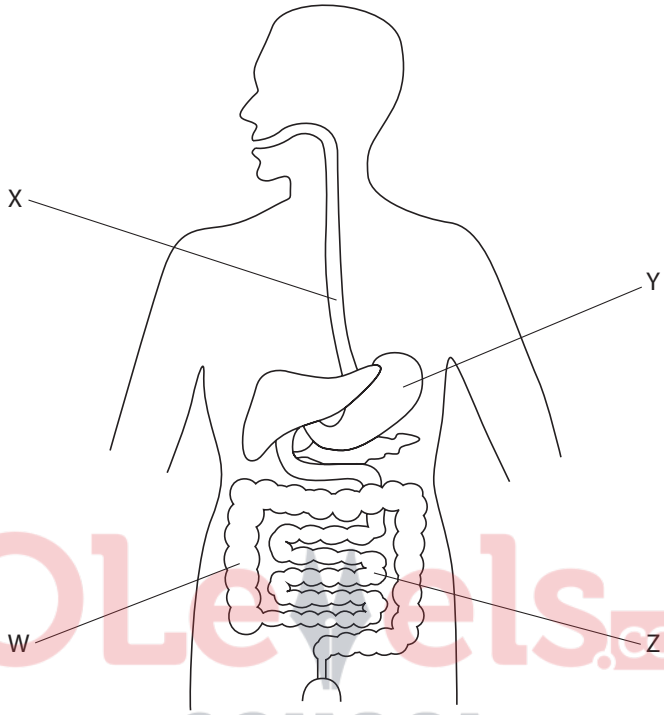
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2. Question 2 · part b.iv · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.iv only.

2 (a) The diagram shows the human digestive system.



(i) Name the structures labelled W, X, Y and Z.

(4)

W .....

X .....

Y .....

Z .....

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2. Question 2 · part b.iv · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.iv only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

relax	coordination	contract	liquid
peristalsis	muscles	enzymes	bolus

Use words from the box to complete the passage.

(4)

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

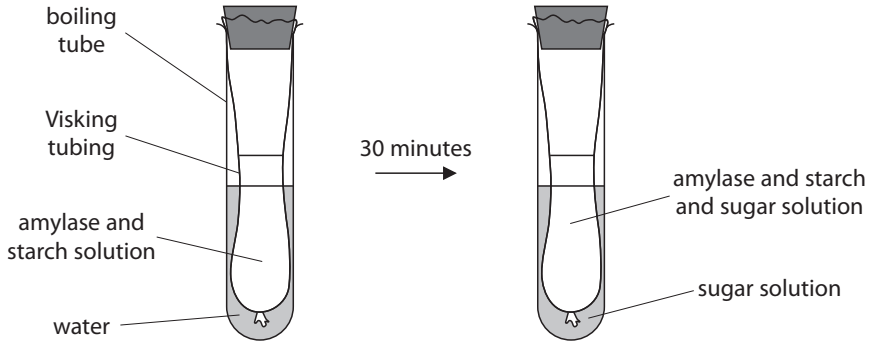
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2. Question 2 · part b.iv · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.iv only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.



(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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2. Question 2 · part b.iv · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.iv only.

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(v) After 30 minutes iodine solution is added to the contents of the Visking tubing.

Explain what the results of this test would show.

(2)

(Total for Question 2 = 15 marks)

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Working and answer

3. Question 5 · part b.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.i only.

5 Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

(3)

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

(1)

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

(1)

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3. Question 5 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 5 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

5 Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

(3)

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

(1)

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

(1)

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4. Question 5 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

(iii) Draw a diagram to show how the student could use the equipment to investigate the effect of glucose concentration on the rate of diffusion.

You should add these labels to your diagram.

- Visking tubing
- boiling tube
- sugar solution
- water

(3)

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4. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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5. Question 3 · part c.i · 1 mark

June 2019 Paper 2

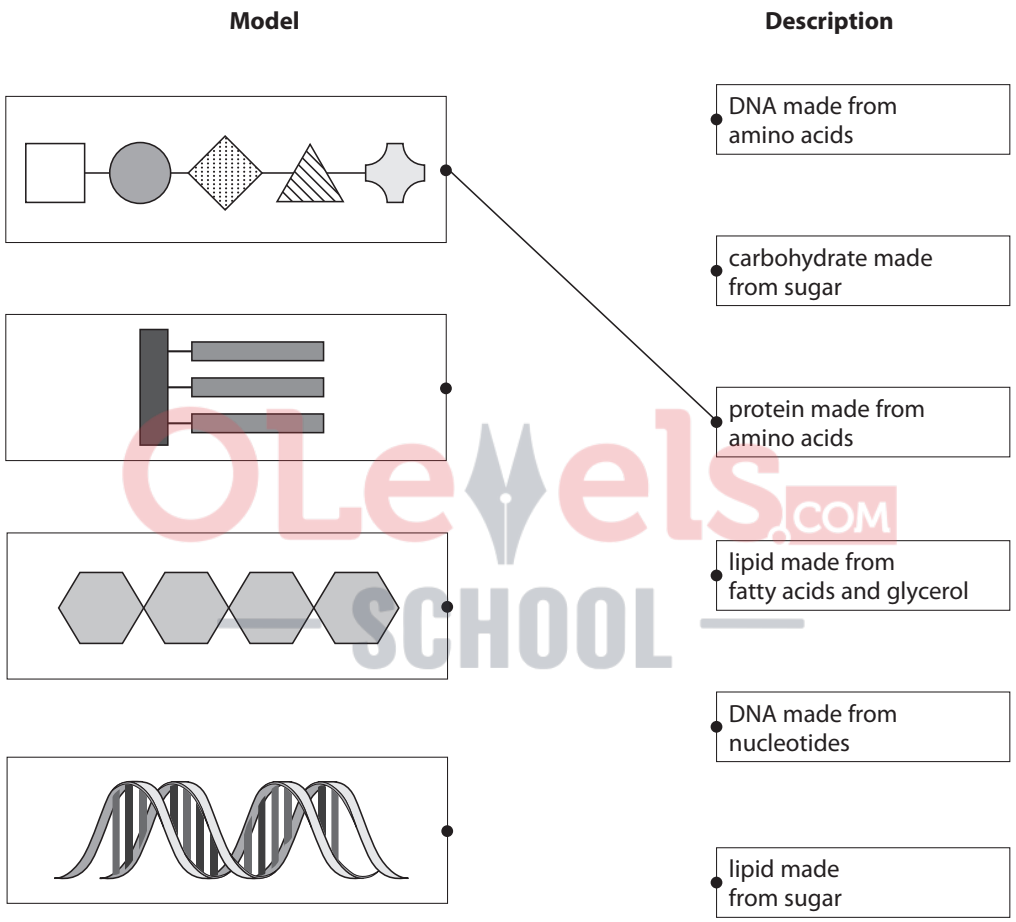
Further practice · Attempt part c.i only.

3 (a) The diagram shows models of four biological molecules.

Draw a straight line from each model to its correct description.

One has been done for you.

(3)



(b) Which body organ produces enzymes that break down protein?

(1)

- ☒ **A** gall bladder
- ☒ **B** large intestine
- ☒ **C** mouth
- ☒ **D** stomach

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5. Question 3 · part c.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part c.i only.

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(c) A student uses this method to test different substances for protein.

- grind solid substances into small pieces
- place substances into separate test tubes
- add a few drops of Biuret reagent to each substance
- record the colour change for each substance

The table shows the student's results.

Substance	Colour after Biuret test
milk	purple
pasta	blue
lemon juice	blue
cheese	purple
distilled water	purple

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(i) In the student's test, how many of the substances give a positive result for protein?

(1)

.....

(ii) The result for distilled water is incorrect.

Suggest one mistake the student could have made to get this incorrect result.

(1)

.....

.....

(d) The table shows an incomplete risk assessment for the Biuret test.

Complete the table by describing how to reduce the risk of each hazard.

(2)

Hazard	Reducing risk
stools – trip hazard	keep stools under bench
broken glass – cuts	
Biuret reagent – irritant	

(Total for Question 3 = 8 marks)

5. Question 3 · part c.i · 1 mark · Working and answer

Working and answer

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6. Question 2 · part b.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.i only.

2 A student uses these three tests to identify the contents of four powders, W, X, Y and Z.

- iodine test
- Benedict's test
- biuret test

The table shows the student's results.

	Colour observed after testing			
Test	**Powder W**	**Powder X**	**Powder Y**	**Powder Z**
iodine	yellow / brown	black	black	yellow / brown
Benedict's	blue	brick red	blue	brick red
biuret	purple	blue	blue	purple

(a) Describe how the student should safely carry out the Benedict's test on the powders.

(4)

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6. Question 2 · part b.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.i only.

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(b) (i) Identify which powder gives a positive result for starch but a negative result for the two other tests.

(1)

.....

(ii) Identify which powders contain protein but no starch.

(2)

.....

.....

(iii) Identify which powder contains reducing sugar but no protein.

(1)

.....

(iv) Give the contents of powder W.

(1)

.....

(Total for Question 2 = 9 marks)

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Working and answer

7. Question 2 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

2 A student uses these three tests to identify the contents of four powders, W, X, Y and Z.

- iodine test
- Benedict's test
- biuret test

The table shows the student's results.

	Colour observed after testing			
Test	**Powder W**	**Powder X**	**Powder Y**	**Powder Z**
iodine	yellow / brown	black	black	yellow / brown
Benedict's	blue	brick red	blue	brick red
biuret	purple	blue	blue	purple

(a) Describe how the student should safely carry out the Benedict's test on the powders.

(4)

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7. Question 2 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

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(b) (i) Identify which powder gives a positive result for starch but a negative result for the two other tests.

(1)

(ii) Identify which powders contain protein but no starch.

(2)

(iii) Identify which powder contains reducing sugar but no protein.

(1)

(iv) Give the contents of powder W.

(1)

(Total for Question 2 = 9 marks)

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Working and answer

8. Question 2 · part b.iii · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.iii only.

2 A student uses these three tests to identify the contents of four powders, W, X, Y and Z.

- iodine test
- Benedict's test
- biuret test

The table shows the student's results.

	Colour observed after testing			
Test	**Powder W**	**Powder X**	**Powder Y**	**Powder Z**
iodine	yellow / brown	black	black	yellow / brown
Benedict's	blue	brick red	blue	brick red
biuret	purple	blue	blue	purple

(a) Describe how the student should safely carry out the Benedict's test on the powders.

(4)

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8. Question 2 · part b.iii · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.iii only.

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(b) (i) Identify which powder gives a positive result for starch but a negative result for the two other tests.

(1)

(ii) Identify which powders contain protein but no starch.

(2)

(iii) Identify which powder contains reducing sugar but no protein.

(1)

(iv) Give the contents of powder W.

(1)

(Total for Question 2 = 9 marks)

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Working and answer

9. Question 2 · part b.iv · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.iv only.

2 A student uses these three tests to identify the contents of four powders, W, X, Y and Z.

- iodine test
- Benedict's test
- biuret test

The table shows the student's results.

	Colour observed after testing			
Test	**Powder W**	**Powder X**	**Powder Y**	**Powder Z**
iodine	yellow / brown	black	black	yellow / brown
Benedict's	blue	brick red	blue	brick red
biuret	purple	blue	blue	purple

(a) Describe how the student should safely carry out the Benedict's test on the powders.

(4)

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9. Question 2 · part b.iv · 1 mark

November 2021 Paper 2

Further practice · Attempt part b.iv only.

DO NOT WRITE IN THIS AREA

(b) (i) Identify which powder gives a positive result for starch but a negative result for the two other tests.

(1)

(ii) Identify which powders contain protein but no starch.

(2)

(iii) Identify which powder contains reducing sugar but no protein.

(1)

(iv) Give the contents of powder W.

(1)

(Total for Question 2 = 9 marks)

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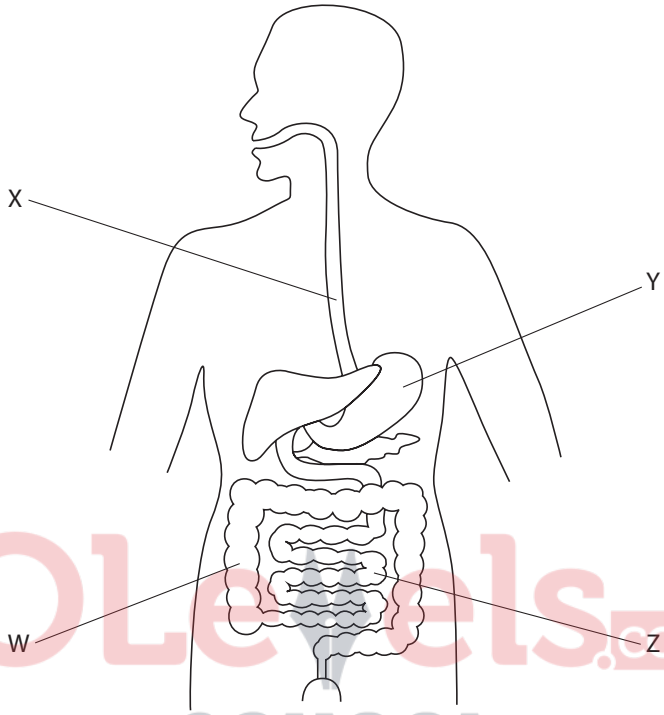
Working and answer

10. Question 2 · part b.v · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.v only.

2 (a) The diagram shows the human digestive system.



(i) Name the structures labelled W, X, Y and Z.

(4)

W .....

X .....

Y .....

Z .....

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10. Question 2 · part b.v · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.v only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

relax	coordination	contract	liquid
peristalsis	muscles	enzymes	bolus

Use words from the box to complete the passage.

(4)

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

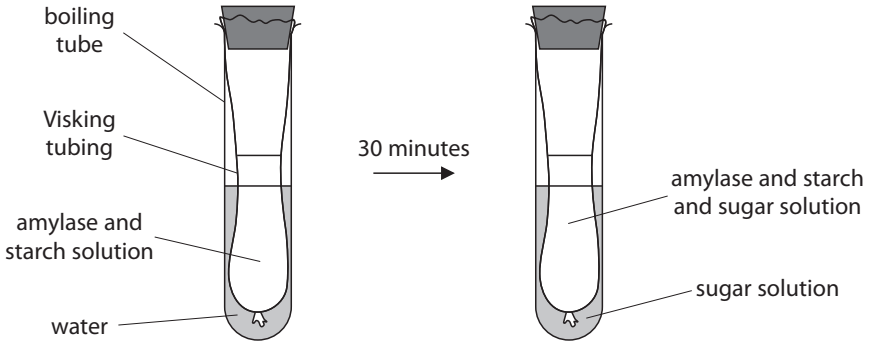
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10. Question 2 · part b.v · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.v only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.



(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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10. Question 2 · part b.v · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.v only.

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(v) After 30 minutes iodine solution is added to the contents of the Visking tubing.

Explain what the results of this test would show.

(2)

(Total for Question 2 = 15 marks)

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Working and answer

11. Question 4 · part a · 1 mark

June 2022 Paper 2

Further practice · Attempt part a only.

4 A student tests foods W, X and Y for the presence of different nutrients.

The student tests each of the foods using three different solutions, A, B and C.

The table shows the student's results.

Food	Solution used in test		
	A	B	C
W	no change	no change	blue-black
X	lilac	no change	no change
Y	no change	brick-red	no change

(a) Name solution C.

(1)

.....

(b) Name the nutrient present in food X.

(1)

.....

(c) Describe a safe method the student could use to test a food using solution B.

(4)

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(Total for Question 4 = 6 marks)

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11. Question 4 · part a · 1 mark · Working and answer

Working and answer

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12. Question 4 · part b · 1 mark

June 2022 Paper 2

Further practice · Attempt part b only.

4 A student tests foods W, X and Y for the presence of different nutrients.

The student tests each of the foods using three different solutions, A, B and C.

The table shows the student's results.

Food	Solution used in test		
	A	B	C
W	no change	no change	blue-black
X	lilac	no change	no change
Y	no change	brick-red	no change

(a) Name solution C.

(1)

.....

(b) Name the nutrient present in food X.

(1)

.....

(c) Describe a safe method the student could use to test a food using solution B.

(4)

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(Total for Question 4 = 6 marks)

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12. Question 4 · part b · 1 mark · Working and answer

Working and answer

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